Exp 10a: Design and Configuration of a Simple Internetwork Using a Router

AIM :

To design and configure a simple internetwork using a router in the Cisco Packet Tracer simulator, assign static IP addresses to the PCs and the router interfaces, and trace how a packet is routed between two different networks.

ALGORITHM :

1. Start the experiment.
2. Open Cisco Packet Tracer and place one router (2911) and two PCs (PC0 and PC1) on the workspace.
3. Connect FastEthernet0 of PC0 to GigabitEthernet0/0 of the router and FastEthernet0 of PC1 to GigabitEthernet0/1 of the router using copper straight-through cables.
4. Open the router CLI and assign IP addresses to its two interfaces (192.168.10.1 and 192.168.20.1, mask 255.255.255.0) and turn them on with `no shutdown`.
5. Assign a static IP address, subnet mask and default gateway to PC0 and PC1 (Desktop → IP Configuration).
6. Send a simple PDU (ICMP) from PC0 to PC1 to check the connectivity.
7. Switch to Simulation mode and trace how the packet travels from PC0 → Router0 → PC1 and the acknowledgment returns from PC1 → Router0 → PC0.
8. Stop the experiment.

PROCEDURE :

A router connects two or more different networks and forwards packets between them. In this network a router and 2 PCs are used; each PC is in a different network (192.168.10.0/24 and 192.168.20.0/24) and the router interface connected to a network acts as the default gateway of the PCs in that network. After forming the network, a simple PDU is transferred from PC0 to PC1 to check the network connectivity.

Step 1 – Configuring Router0: Select the router, open the CLI, press ENTER to start the configuration and type `enable` to activate the privileged mode.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#end
Router#
```

Step 2 – Configuring the PCs: Select the PC, go to Desktop → IP Configuration and assign the IP address, subnet mask and default gateway. The default gateway of PC0 is 192.168.10.1 and that of PC1 is 192.168.20.1.

Step 3 – Connecting the PCs with the router: Connect the FastEthernet0 port of PC0 with GigabitEthernet0/0 of Router0 and the FastEthernet0 port of PC1 with GigabitEthernet0/1 of Router0 using copper straight-through cables.

Router configuration table:

Device	Interface	IP address	Subnet mask
Router0	GigabitEthernet0/0	192.168.10.1	255.255.255.0
Router0	GigabitEthernet0/1	192.168.20.1	255.255.255.0

PC configuration table:

Device	IP address	Subnet mask	Default gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.20.2	255.255.255.0	192.168.20.1

Testing and simulation: Click the *Add Simple PDU* tool, click PC0 and then PC1. In Simulation mode press *Auto Capture / Play* and observe the ICMP packet moving from PC0 to the router and from the router to PC1, followed by the acknowledgment (reply) from PC1 back to PC0 through the router. The status “Successful” in the PDU list shows that the internetwork works.

OUTPUT :



Fig 1. Designed network topology – PC0 and PC1 connected through Router0 (2911)

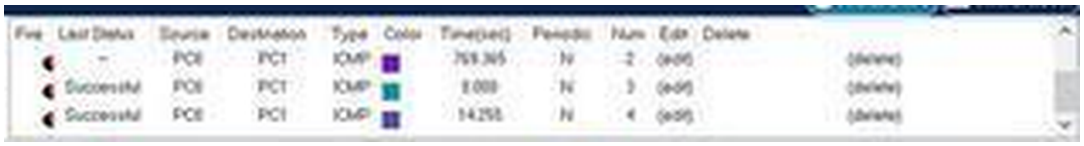


Fig 2. PDU list – ICMP PDUs sent from PC0 to PC1 through the router (Last Status: Successful)

RESULT :

Thus, a simple internetwork was designed and configured using a router in Cisco Packet Tracer. Static IP addresses were assigned to the PCs and the router interfaces, and the packet transmission from PC0 to PC1 through the router was successfully verified.